PdfHelperReader – Comprehensive User Manual

High-Performance Dual-Pane Reading, Synced Auto-Scroll & Instant Vocabulary Generation

1. Welcome & Introduction

PdfHelperReader is a specialized Android reading environment designed for language learning, academic research, and deep study. It pairs your main book document with a companion vocabulary sidecar PDF or an embedded Google Web Search browser in real-time.

2. Upper Main Ribbon Controls (Top Action Bar)

Button	Function & Detailed How-To-Use Guide
B	**Add Bookmark**: Tap to bookmark your current reading position. Captures creation timestamp and extracts the first 25 words from the book page.
BL	**Bookmark List**: Opens saved bookmarks dialog. Shows creation date/time and 25-word book page snippet previews. Tap any item to jump directly to both book and vocab pages.
Both/Book/Vocab	**Split Mode Toggle**: Cycles screen layout between *Both* (dual-pane split view), *Book* (full-screen main document), or *Vocab* (full-screen vocabulary pane).
Z	**Zen Mode**: Hides all toolbars for distraction-free reading, replacing them with a docked vertical right-edge control strip.
SR	**Synced Reading**: Toggles 1:1 real-time bidirectional scroll synchronization. When scrolling either the Book or Vocab pane, the other pane scrolls automatically in lockstep.
AS	**Auto Scroll**: Gentle continuous frame-by-frame auto-scroll calibrated at 1 viewport height / 5 minutes. Touch dragging is supported without stopping auto-scroll.
Sync	**Page Sync**: Instantly jumps the Vocabulary pane to match your current Book page using the exact *vocab_map* array.
G:Vocab / G:Google	**Vocab Mode**: Toggles lower pane between the companion Vocabulary PDF sidecar and the embedded Google Web Search browser.
T	**Theme Toggle**: Switches Dark/Light mode with high-contrast ColorMatrix PDF page color inversion for comfortable night reading.
R:C / R:P	**Reading Mode**: Toggles Continuous vertical scrolling (*R:C*) or fit-to-width Page-by-Page mode (*R:P*).
V:PDF / V:TXT	**View Mode**: Toggles vector PDF page rendering with transparent OCR text selection overlay (*V:PDF*) or plain text extraction (*V:TXT*).
H	**History**: Opens reading history showing First Opened & Last Closed timestamps, with a *Del vocab* button to delete associated vocabulary files.
O	**OCR Generator**: Applies Google ML Kit OCR to image-based or scanned PDFs, adding a searchable text layer saved to *Downloads/PdfHelperReader_files/*.
OB / OV / LV	**Open Book / Open Vocab / Load Vocab**: Open book/vocab files via system file picker, or generate vocabulary PDF automatically.

3. Per-Pane Navigation & Header Controls (PaneBar)

• **Header Navigation (Left)**: Prev/Next page buttons, Page Jump Counter (e.g. *18 / 512*), and Zoom (- / 100% / +) are docked on the left. Tap page counter to open the **Go to Page** jump dialog.

• **Document Title (Right)**: Displays book or vocab title. Horizontally scrollable—swipe left/right across the title to read long file names. Tap title to view document info.

4. Zen Mode Controls (Right-Edge Strip)

When Zen Mode (**Z**) is active, all toolbars are hidden. A sleek vertical control strip docks against the right screen edge containing:

• **Page Number Badge**: Displays current page (e.g. *18*). Tap to open Page Jump dialog directly.

• **B**: Add bookmark instantly.

• **G**: Toggle lower pane between Vocab PDF and Google Web Search.

• **Both / Book / Vocab**: Cycle view modes directly from Zen mode.

• **X**: Exit Zen mode and restore toolbars.

5. Instant 2-Page Temporary Vocabulary Generation

When generating a vocabulary PDF for a book for the first time:

1. **Instant 2-Page Preview**: The app calculates a 2-page window *[current_page .. current_page + 1]* and generates a temporary vocab PDF in **~0.1 seconds**.

2. **Immediate Display**: The temporary vocab PDF loads into the lower pane immediately so you can start reading definitions without waiting.

3. **Background Full Build & Auto-Jump**: The full vocabulary PDF generates for all pages in the background. Once finished, it saves to *Downloads/PdfHelperReader_files/*, auto-jumps to *vocab_map[current_page]*, and deletes the temporary preview file.

6. Text Selection & Google Web Search Integration

• Highlight any word or phrase on PDF pages in **V:PDF** mode using native touch selection.

• A floating search banner appears (*Search "..." in Google?*). Tapping **Google** switches to dual-pane mode and launches Google Web Search in the lower pane.

• **Chrome-Style Web Bar**: Google Web mode features **‹ Back**, **Forward ›**, **■ Reload**, and page title controls.

• **System Back Gesture Interception**: Doing a back gesture in Google mode navigates backward in web search history instead of exiting the app. At the root search page, back returns you to the Vocab PDF pane.

7. Dictionary Database & Memory Optimization

• **1.8 Million Entry Dictionary**: Merged from Webster's English Dictionary and WordNet into 133,273 root words and 1.83M variant keys using *0$* reference encoding.

• **3 Short Meanings**: Definitions are formatted into up to 3 short, easy-to-understand meanings.

• **Inflection Variants**: Plurals, past-tense verbs, and gerunds (e.g. *split*, *splits*, *splitted*, *splitting*) resolve instantly to the exact same 3-meaning definition.